How to Design Mechanical Keyboard PCBs with Kicad

At NextPCB, we're proud to support innovative makers like Joe Scotto, a mechanical keyboard enthusiast and engineer who shares his passion through hands-on DIY projects. His latest tutorial, "Designing Mechanical Keyboard PCBs with KiCad," is a fantastic resource for anyone looking to build a custom keyboard from scratch. Thanks to his detailed guidance and exclusive KiCad library, even beginners can dive into PCB design with confidence.

Project Overview: A 3×3 Macro Pad

Joe's tutorial focuses on creating a 3×3 macro pad—a compact yet functional keyboard that teaches core concepts like:

- Keyboard matrix wiring (rows, columns, and diodes)
- PCB layout in KiCad (switch placement, trace routing)
- Microcontroller integration (using an Arduino Pro Micro)

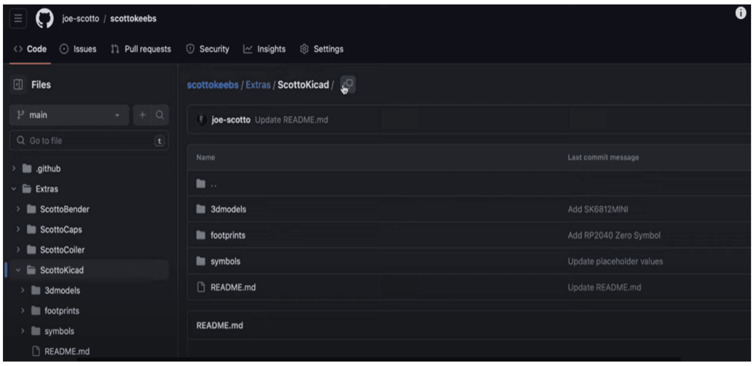
This project is perfect for learning PCB design fundamentals before tackling larger keyboards.

Key Steps in the Build Process

1. KiCad Setup & Custom Libraries

Joe provides a dedicated KiCad library with pre-made symbols and footprints for switches, diodes, and controllers, speeding up the design process1.

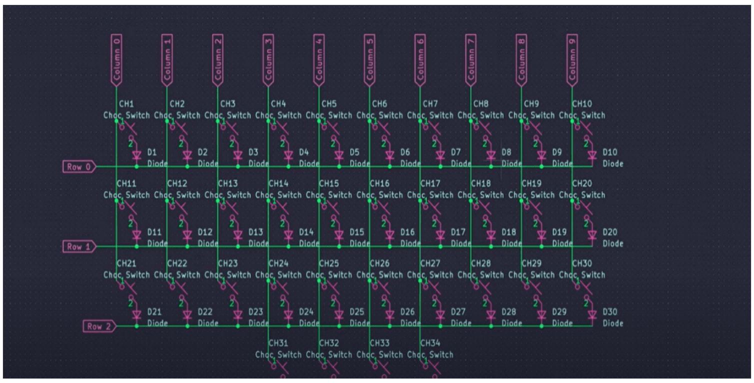
A Gerber export plugin simplifies file generation for manufacturing.



2. Schematic Design

The keyboard matrix is built with switches and diodes (to prevent ghosting).

Global labels keep wiring clean and organized.



3. PCB Layout & Routing

- Switches are placed on a 19.05mm grid (standard MX spacing).
- Diodes are flipped to the PCB's backside for a cleaner look.
- Dual-layer routing: Rows on the front, columns on the back, with vias for layer transitions.

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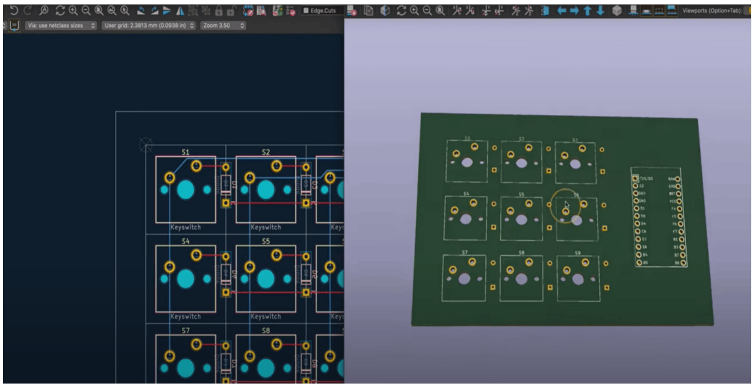

4. Final Checks & Production

- Design Rule Check (DRC) ensures no errors.
- A board outline (Edge Cuts layer) is added for fabrication.
- Gerber files are exported for production—NextPCB offers high-quality, affordable PCB services for such projects!

Watch the Full Tutorial

For a visual walkthrough, check out Joe's YouTube tutorial. Whether you're building a macro pad or a full-sized keyboard, this guide is your gateway to custom keyboard design! You can find the ScottoKicad library here. Follow Joe Scotto's work at scottokeebs.com and explore more keyboard projects!

Ready to start your project? Upload your Gerber files to NextPCB for reliable, high-quality fabrication. Share your builds with us—we'd love to see what you create!

Still, need help? Contact Us: support@nextpcb.com

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